Cantor Set as a Fractal and Its Application in Detecting Chaotic Nature of Piecewise Linear Maps

by

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DECLARATION

I hereby declare that the work presented in this dissertation, titled **"Cantor Set as a Fractal and Its Application in Detecting Chaotic Nature of Piecewise Linear Maps"**, is the result of my independent research conducted at the Department of Mathematics, St. Xavier's College (Autonomous), Kolkata, in partial fulfillment of the requirements for the degree of Bachelor of Science (Honors) in Mathematics.

This dissertation was carried out under the guidance of Dr.Pabitra Debnath, and to the best of my knowledge, it has not been submitted elsewhere for any other degree or academic qualification.I do not claim the entire work to be my own. However, some theorems and many of the results proved in this project comprise original contributions. I have sincerely endeavored to uphold academic ethics and honesty. I affirm that I have cited all external sources, and no part of my dissertation includes unacknowledged materials.

Debkanta Ghosh

ABSTRACT

In this dissertation we have explored the Cantor set through the lens of fractals and box-counting dimension. The Cantor set can be geometrically constructed by the continuous and infinite removal of segments from the closed unit interval [0, 1]. The remaining points in the unit interval after this iterative removal process constitute the Cantor set. Notably, the dimension of this set is not an integer; rather, it possesses a 'fractional' dimension, which classifies it as a fractal. The Cantor set exemplifies an uncountable set with a measure of zero and has significant implications in various fields of mathematics, including topology, measure theory, dynamical systems, and fractal geometry. In this paper, we present three types of generalizations of the Cantor set based on different removal processes. Additionally, we examine certain characteristics of the fractal dimensions associated with these generalized Cantor sets. Furthermore, we demonstrate its application in identifying the chaotic nature of dynamics generated by the iteration of a piecewise linear map.

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1 Introduction

In the late nineteenth century, the German mathematician George Cantor (1845–1918) discovered the Cantor set. In 1883, he published a series of papers in which he gave the first systematic treatment of the point-set topology of the real line, opening several issues and capturing the attention of researchers in the field of set theory. One of these contributions, which he presented as an example of a perfect, nowhere dense subset of R is the classical Cantor set. This caused quite a stir in the mathematical community, as the properties of the Cantor set were remarkably unintuitive. For example, points between the removed intervals are weeded out in the construction of the Cantor set, making it difficult to envision points in the set other than the endpoints of the removed intervals. Yet, the existence of such points is an implication of the construction process. In contemporary language, the Cantor set is a classic example of a perfect, closed subset of the interval $[0, 1]$ that has the same cardinality as the real line while possessing a Lebesgue measure of zero. For much of the twentieth century, the Cantor set was regarded as a mere mathematical curiosity, but Stephen Smale's work in the 1960s showed that Cantor sets play a crucial role in dynamical systems.

Although Euclidean geometry is a useful mathematical tool for describing simple objects with ideal shapes such as cubes, cones, and cylinders, most objects in the real world are so complex and irregular that their shapes and dimensions cannot be described by classical Euclidean geometry. Examples include the motion of a particle suspended in a fluid (Brownian motion), the length of a coastline, the surface area of the human lung, and the shapes of trees and clouds. The study of such objects led to the development of a new area in mathematics known as fractal geometry. The mathematician Benoit Mandelbrot popularized the concept of fractal geometry and first coined the term in 1977. Computer simulations of complex maps of the complex plane he created have produced exceedingly intricate and beautiful fractals.

A fractal is a geometric figure with a complex shape in which details appear similar at different spatial scales. Fractals are self-similar, meaning that subsets of a fractal resemble the entire fractal. As a result, measuring the area of a fractal is typically challenging. A fractal can be characterized using a fractal dimension, which has been applied in texture analysis and texture segmentation.

Earthquakes are a dynamic stick-slip process occurring at fault lines. The Earth's solid crust consists of approximately 12 tectonic plates that shift around 2–3 centimeters per year. When the stress on a fault exceeds a threshold, rock fracturing occurs, depending on the sticking strength of the fault, and a portion of the crust slips, releasing potential elastic energy in the form of earthquakes. These 'quakes' can be modeled as accumulated overlaps of two Cantor sets moving with uniform relative velocity. Seismic clustering and its quantitative analysis have been studied using the Cantor set model. A primary goal

of this investigation is to test whether variations in the fractal dimension of earthquake clustering might serve as a precursor to major earthquakes. Additionally, the Cantor set fractal distribution and probability concepts in the time domain have been employed to estimate recurrence periods for earthquakes through certain threshold levels of magnitude, as studied by Sezer.

Several definitions of fractal dimension are useful in different contexts. Various approaches have been suggested to determine an appropriate dimension for the object under investigation. The box-counting dimension is the most common method of measurement in many applied fields due to its simplicity and automatic computability. Researchers have proposed multiple algorithms for calculating fractal dimension using the box-counting scheme.

The Cantor set, initially formed for pure mathematical exploration, has recently been found to serve as an ideal model for a wide range of real-world phenomena—from strange attractors in nonlinear dynamical systems to the distribution of galaxies in the universe. Today, the Cantor set holds a celebrated place in mathematical analysis and its applications.

2 Construction of the Cantor Middle One-Third Set

To construct the Cantor middle one-third set C, we begin with the closed interval:

$$F_0 = I = \{x \in \mathbb{R} : 0 \leq x \leq 1\},$$

where $\mathbb{R}$ is the set of real numbers. First, we trisect F_0 and then remove the middle open one-third interval:

$$D_1 = \left\{x \in \mathbb{R} : \frac{1}{3} < x < \frac{2}{3}\right\}.$$

We get the remaining set as:

$$F_1 = F_0 - D_1 = \left\{x \in \mathbb{R} : 0 \leq x \leq \frac{1}{3}\right\} \cup \left\{x \in \mathbb{R} : \frac{2}{3} \leq x \leq 1\right\}.$$

The sets F_0 and F_1 are termed, respectively, as the initiator and generator in the process of constructing the Cantor middle one-third set C. Clearly, F_1 is the union of two closed intervals, each of length $\frac{1}{3}$.

Again, from each of these two closed intervals, we remove the respective middle open intervals of length $\frac{1}{9}$, that is,

$$D_2 = \left\{x \in \mathbb{R} : \frac{1}{9} < x < \frac{2}{9}\right\} \cup \left\{x \in \mathbb{R} : \frac{7}{9} < x < \frac{8}{9}\right\}.$$

After this removal, we get a smaller set that is the union of four closed intervals of length $\frac{1}{9}$, which we term as F_2, i.e.,

$$\begin{aligned}
&\mathrm{F}_2 = F_0 - (D_1 \cup D_2) \\
&= \left\{x \in \mathbb{R} : 0 \leq x \leq \tfrac{1}{9}\right\} \cup \left\{x \in \mathbb{R} : \tfrac{2}{9} \leq x \leq \tfrac{3}{9}\right\} \\
&\cup \left\{x \in \mathbb{R} : \tfrac{6}{9} \leq x \leq \tfrac{7}{9}\right\} \cup \left\{x \in \mathbb{R} : \tfrac{8}{9} \leq x \leq 1\right\}.
\end{aligned}$$

Continuing this way, in the nth step, we remove a set D_n, which is the union of 2^{n-1} disjoint open intervals, each of length 3^{-n}. After this removal, we get the set F_n, which is the union of 2^n disjoint closed intervals of length 3^{-n} each.

Definition 2.1 The Cantor middle one-third set C is defined as:

$$C = I - \bigcup_{n=1}^{\infty} D_n = \bigcap_{n=1}^{\infty} F_n,$$

where F_{n+1} is constructed as above, i.e., by trisecting F_n and removing the open middle one-third interval, with F_0 being the closed real interval $[0, 1]$.

Lemma 2.1 A number x in the closed interval $[0, 1]$ belongs to the Cantor middle one-third set C if and only if the triadic expansion of x can be expressed without the digit '1'.

In view of the above proposition, one can define Cantor middle one-third set symbolically as follows:

Definition 2.2 The Cantor middle one-third set C is the set of those real numbers x which can be represented as:

$$x = \sum_{n=0}^{\infty} a_n 3^{-n},$$

where $a_n \in \{0, 2\}$ for all $n \geq 0$.

Lemma 2.2 The Cantor middle one-third set C is closed, bounded, and non-empty.

Lemma 2.3 The Cantor set C is uncountable.

Lemma 2.4 The Cantor set C is compact.

Defination 2.3 A subset A of the real line $\mathbb{R}$ is said to be totally disconnected (or has empty interior), if A contains no non-empty open intervals.

Lemma 2.5 The Cantor middle one-third set C is totally disconnected .

Definition 2.4 A subset A of the set of real numbers $\mathbb{R}$ is called a perfect set if every point of A is a limit point of A.

Lemma 2.6 The Cantor middle one-third set C is perfect.

Theorem 2.1 Any subset E of the real line $\mathbb{R}$ which is compact, totally disconnected, and perfect is homeomorphic to the Cantor middle one-third set C.

In view of Theorem 2.1, one can define a Cantor set as follows:

Definition 2.5 Any subset of the real line $\mathbb{R}$ which is compact, totally disconnected, and perfect is called a Cantor-like set or simply a Cantor set.

3 Concept of Fractal Dimension

The Cantor set is the prototype of a fractal. It is an object which appears self-similar under varying degrees of magnification. One of the typical features of a fractal is its fractal dimension, which is essentially a measure of self-similarity. It is sometimes referred to as similarity dimension.

There are a number of non-equivalent ways of defining fractal dimension. One of the most popular ways of computing the dimension of a set is to find the box-counting dimension of it, which is defined as follows:

Definition 3.1 Let $N_{\text{box}}(K, r)$ be the minimum number of axis-parallel square boxes of side length r needed to cover the set K. The box-counting dimension $D_B(K)$ of the set K is defined as:

$$D_B(K) = \lim_{r \to 0} \frac{\log N_{\text{box}}(K, r)}{\log(1/r)}.$$

For example, in the nth stage of the construction of the Cantor middle one-third set C, we need 2^n boxes of length 3^{-n} each. Therefore,

$$D_B(C) = \lim_{n \to \infty} \frac{\log 2^n}{\log 3^n} = \frac{\log 2}{\log 3}.$$

4 Generalization of Cantor Middle One-Third Set

In this section, an attempt has been made to construct a generalized Cantor set. Depending upon the removal process of a portion of the closed unit interval $[0, 1]$ in the construction of the Cantor set, we consider three types of generalization.

4.1 Generalization I

Let us consider the closed interval $F_0 = [0, 1]$. Now divide F_0 into $n \geq 3$ equal parts and remove from it the central open interval of length $1 - \frac{2}{n}$, i.e., we are removing:

$$D_1 = \left\{ x \in \mathbb{R} : \frac{1}{n} < x < 1 - \frac{1}{n} \right\}.$$

From $F_0 = [0, 1]$, we get the union of two closed intervals:

$$F_1 = F_0 - D_1 = \left[0, \frac{1}{n}\right] \cup \left[1 - \frac{1}{n}, 1\right] = L \cup R,$$

where $L = \left[0, \frac{1}{n}\right]$ and $R = \left[1 - \frac{1}{n}, 1\right]$.

Next, we divide each of L and R into n equal parts and remove from them the central open intervals $\left(\frac{1}{n^2}, \frac{n-1}{n^2}\right)$ and $\left(1 - \frac{n-1}{n^2}, 1 - \frac{1}{n^2}\right)$, respectively, i.e.,

$$D_2 = \left\{ x \in \mathbb{R} : \frac{1}{n^2} < x < \frac{n-1}{n^2} \right\} \cup \left\{ x \in \mathbb{R} : 1 - \frac{n-1}{n^2} < x < 1 - \frac{1}{n^2} \right\}.$$

The remaining part F_2 is the union of four closed intervals:

$$F_2 = F_0 - (D_1 \cup D_2) = LL \cup LR \cup RL \cup RR,$$

where

$$\begin{aligned}
LL &= \left[0, \frac{1}{n^2}\right], \\
LR &= \left[\frac{n-1}{n^2}, \frac{1}{n}\right], \\
RL &= \left[1 - \frac{1}{n}, 1 - \frac{n-1}{n^2}\right], \\
RR &= \left[1 - \frac{1}{n^2}, 1\right].
\end{aligned}$$

Continuing this way, we get a sequence of closed intervals $\{F_k\}$, where $F_0 = [0,1]$ contains one closed interval of length 1, F_1 contains 2 closed intervals each of length $\frac{1}{n}$, F_2 contains 2^2 closed intervals each of length n^{-2}, i.e., in general, F_k contains 2^k closed intervals of length n^{-k} each.

The limiting set formed by this process is called the Cantor $\left(1-\frac{2}{n}\right)$ th-set, and we denote it by:

$$C(1-\frac{2}{n}), \quad (n \geq 3),$$

i.e.,

$$C(1-\frac{2}{n}) = F_0 - \bigcup_{i=1}^{\infty} D_i = \bigcap_{k=0}^{\infty} F_k.$$

4.2 Generalization II

In this generalization, we construct the Cantor $\left(\frac{1}{n}\right)$th set by removing the central open interval of length $\frac{1}{n}$ from $'n'$ $(n \geq 3)$ equal parts of the unit interval $[0,1]$.

Let $F_0 = [0,1]$. We divide F_0 into $n \geq 3$ equal parts and remove the central open interval of length $\frac{1}{n}$, denoted as:

$$D_1 = \left\{x : \frac{n-1}{2n} < x < \frac{n+1}{2n}\right\}$$

The remaining set is:

$$F_1 = F_0 - D_1 = \left[0, \frac{n-1}{2n}\right] \cup \left[\frac{n+1}{2n}, 1\right] = L \cup R$$

Next, we divide each of L and R into n equal parts and remove the central open intervals:

$$D_2 = \left\{x : \frac{n^2-2n+1}{4n^2} < x < \frac{n^2-1}{4n^2}\right\} \cup \left\{x : \frac{3n^2+1}{4n^2} < x < \frac{3n^2+2n-1}{4n^2}\right\}$$

The remaining set is:

$$F_2 = F_0 - (D_1 \cup D_2) = \text{LL} \cup \text{LR} \cup \text{RL} \cup \text{RR}$$

where:

$$\text{LL} = \left[0, \frac{n^2-2n+1}{4n^2}\right], \quad \text{LR} = \left[\frac{n^2-1}{4n^2}, \frac{n-1}{2n}\right]$$

$$\text{RL} = \left[\frac{n+1}{2n}, \frac{3n^2+1}{4n^2}\right], \quad \text{RR} = \left[\frac{3n^2+2n-1}{4n^2}, 1\right]$$

Continuing this process, at the k-th step, F_k consists of 2^k closed intervals, each of length $\left(\frac{n-1}{2n}\right)^k$. The limiting set is:

$$C\left(\frac{1}{n}\right) = \bigcap_{k=0}^{\infty} F_k = F_0 - \left(\bigcup_{i=1}^{\infty} D_i\right)$$

4.3 Generalization III

In this generalization, we divide the unit interval $F_0 = [0, 1]$ into n $(n \geq 3)$ equal parts, where n is an odd number, say $n = 2m + 1$. We then remove from it the alternate m open intervals, viz.

$$\left(\frac{1}{n}, \frac{2}{n}\right), \left(\frac{3}{n}, \frac{4}{n}\right), \ldots, \left(\frac{2m-1}{n}, \frac{2m}{n}\right).$$

That is, we remove the set:

$$D_1 = \bigcup_{k=1}^{m} \left(\frac{2k-1}{n}, \frac{2k}{n}\right).$$

After removing D_1, the remaining set F_1 is the union of $(m + 1)$ closed intervals:

$$F_1 = F_0 - D_1 = \left[0, \frac{1}{n}\right] \cup \left[\frac{2}{n}, \frac{3}{n}\right] \cup \cdots \cup \left[\frac{2m}{n}, 1\right].$$

This serves as the generator of our proposed Generalization III.

Next, we subdivide each of these $(m+1)$ closed intervals into n equal parts and remove from each of them the 2nd, 4th, ..., $2m$-th open intervals. Continuing this process, we get a sequence $\{F_k\}$ of closed intervals:

- One interval of length 1 in step 0,
- $(m + 1)$ intervals of length n^{-1} in step 1,
- $(m + 1)^2$ intervals of length n^{-2} in step 2,
- In general, $(m + 1)^k$ closed intervals of length n^{-k} in step k.

The limiting set obtained in this process is called the **Cantor $\frac{m}{n}$-th set**, denoted by $C(\frac{m}{n})$. Formally,

$$C(\frac{m}{n}) = \bigcap_{k=0}^{\infty} F_k.$$

4.4 Properties of Generalized Cantor Sets

Theorem 4.1. The sets $C(\frac{n-2}{n})$, $C(\frac{1}{n})$, and $C(\frac{m}{n})$ are all compact, totally disconnected, and perfect subsets of $F_0 = [0, 1]$.

Proof. First, the sets $C(\frac{n-2}{n})$, $C(\frac{1}{n})$, and $C(\frac{m}{n})$ are all subsets of the closed interval $F_0 = [0, 1]$, making them bounded. They are also closed, as an infinite intersection of closed sets remains closed. By the Heine-Borel theorem, they are compact, being closed and bounded subsets of $\mathbb{R}$.

Consider the set $C(\frac{n-2}{n})$, which satisfies the recurrence relation:

$$\lambda(F_k) = \frac{2}{n}\lambda(F_{k-1}), \quad k = 1, 2, 3, \ldots,$$

where $\lambda(F_k)$ denotes the Lebesgue measure of the closed interval F_k. Since $\lambda(F_0) = 1$, for $n \geq 3$, we get:

$$\lambda(C(\frac{n-2}{n})) = \lim_{k\to\infty} \lambda(F_k) = 0.$$

Thus, $C(\frac{n-2}{n})$ does not contain any non-empty intervals, proving that it is totally disconnected. Similarly, we establish total disconnectedness for the other two sets with similar recurrence relations:

(i) For $C(\frac{1}{n})$:

$$\lambda(F_k) = \frac{n-1}{n}\lambda(F_{k-1}), \quad k = 1, 2, 3, \ldots$$

(ii) For $C(\frac{m}{n})$:

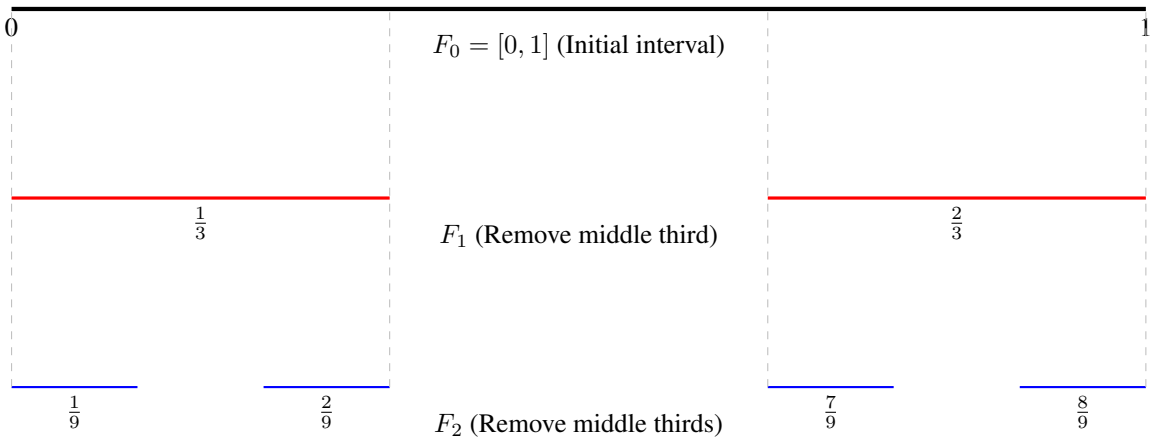
$$\lambda(F_k) = \frac{m+1}{n+1}\lambda(F_{k-1}), \quad k = 1, 2, 3, \ldots, \quad \text{where } n = 2m.$$

Finally, to prove that $C(\frac{n-2}{n})$ is perfect, let x_0 be any arbitrary point of $C(\frac{n-2}{n})$. For any given $\epsilon > 0$, we choose a sufficiently large natural number k such that $n^{-k} < \epsilon$. Then, the sequence $\{c_k\}$ of points in $C(\frac{n-2}{n})$ within the closed interval F_k is infinite and contained in the open interval $(c_0 - \epsilon, c_0 + \epsilon)$. Since $|c_k - x_0| < \epsilon$ as $k \to \infty$, it follows that x_0 is a limit point of $C(\frac{n-2}{n})$. Since x_0 is arbitrary, $C(\frac{n-2}{n})$ is a perfect set. The same argument applies to $C(\frac{1}{n})$ and $C(\frac{m}{n})$. □

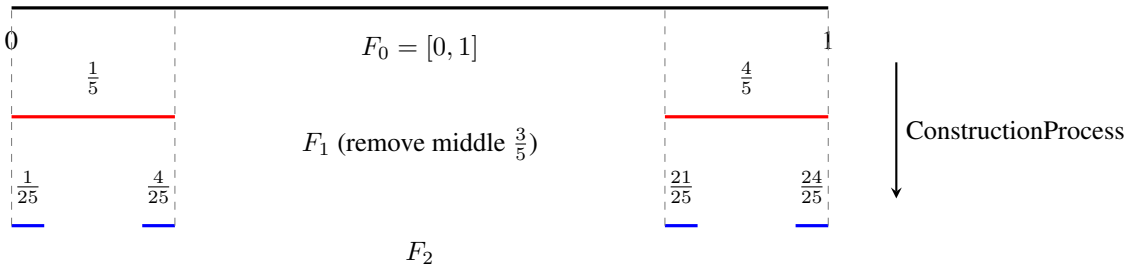
Remark 4.1. By Definition 2.5 and Theorem 4.1, we conclude that the constructed sets $C(\frac{n-2}{n})$, $C(\frac{1}{n})$, and $C(\frac{m}{n})$ are all Cantor sets. □

Cantor Set Construction

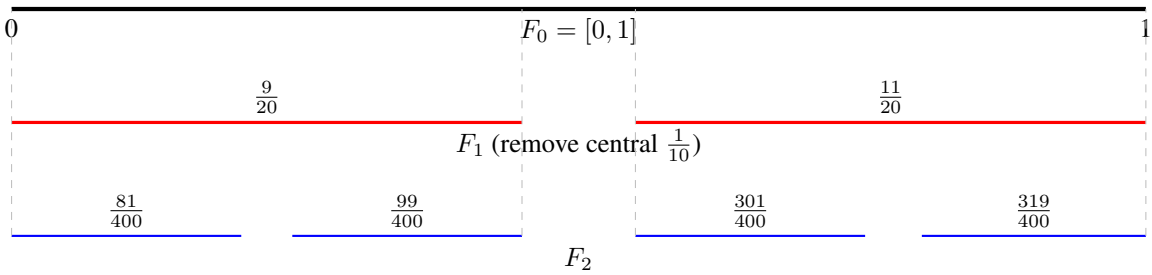
$F_0 \supset F_1 \supset F_2 \supset \cdots \supset C$



Construction of Generalized Cantor Set $C\left(\frac{3}{5}\right)$



Construction of $C\left(\frac{1}{10}\right)$ **Set**



5 Dimension and Its Characteristics

To compute the dimension of the sets $C\left(\frac{n-2}{n}\right)$, $C\left(\frac{1}{n}\right)$ and $C\left(\frac{m}{n}\right)$, we proceed as follows:

For the set $C\left(\frac{n-2}{n}\right)$, we have:

$$N_{\text{box}}\left(C\left(\frac{n-2}{n}\right), n^{-k}\right) = 2^k; \quad k = 0, 1, 2, \ldots$$

Hence:

$$D_B\left(C\left(\frac{n-2}{n}\right)\right) = \lim_{k\to\infty} \frac{\log 2^k}{\log n^k} = \frac{\log 2}{\log n}; \quad n = 3, 4, 5, \ldots$$

For the set $C\left(\frac{1}{n}\right)$, we have:

$$N_{\text{box}}\left(C(1/n), \left(\frac{n-1}{2n}\right)^k\right) = 2^k; \quad k = 0, 1, 2, \ldots$$

and therefore:

$$D_B(C(1/n)) = \lim_{k\to\infty} \frac{\log 2^k}{\log\left(\frac{2n}{n-1}\right)^k} = \frac{\log 2}{\log\left(\frac{2n}{n-1}\right)}; \quad n = 3, 4, 5, \ldots$$

For the set $C\left(\frac{m}{n}\right)$, we have:

$$N_{\text{box}}\left(C\left(\frac{m}{n}\right), n^{-k}\right) = (m+1)^k; \quad k = 0, 1, 2, 3, \ldots$$

Thus:

$$D_B\left(C\left(\frac{m}{n}\right)\right) = \lim_{k\to\infty} \frac{\log(m+1)^k}{\log n^k} = \frac{\log(1+m)}{\log(1+2m)}; \quad m = 1, 2, 3, \ldots$$

Remark 5.1. The generator of $C(\frac{n-2}{n})$ for $n = 3$ is:

$$F_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right],$$

which is the generator of the classical Cantor middle one-third set C.

Similarly, the generators of the sets $C(\frac{1}{n})$ and $C(\frac{m}{n})$ for $n = 3$ are also equivalent to the generator of the Cantor middle one-third set C.

Furthermore, the dimensions of $C(\frac{n-2}{n})$, $C(\frac{1}{n})$, and $C(\frac{m}{n})$ for $n = 3$ all coincide with the dimension of the Cantor middle one-third set C. This allows us to conclude that $C(\frac{n-2}{n})$, $C(\frac{1}{n})$, and $C(\frac{m}{n})$ are all generalizations of the Cantor middle one-third set C. □

Theorem 5.1:

The sequence $D_B(C(\frac{n-2}{n}))$ of dimensions of the family of Cantor sets

$$C(\frac{n-2}{n}), \quad n = 3, 4, 5, \ldots$$

is a decreasing function of n, and it asymptotically approaches zero.

Proof: For the set $C(\frac{n-2}{n})$, we have

$$D_B(C(\frac{n-2}{n})) = \frac{\log 2}{\log n}, \quad n \geq 3.$$

Differentiating with respect to n, we obtain

$$\frac{d}{dn} D_B(C(\frac{n-2}{n})) = -\frac{\log 2}{n(\log n)^2} < 0.$$

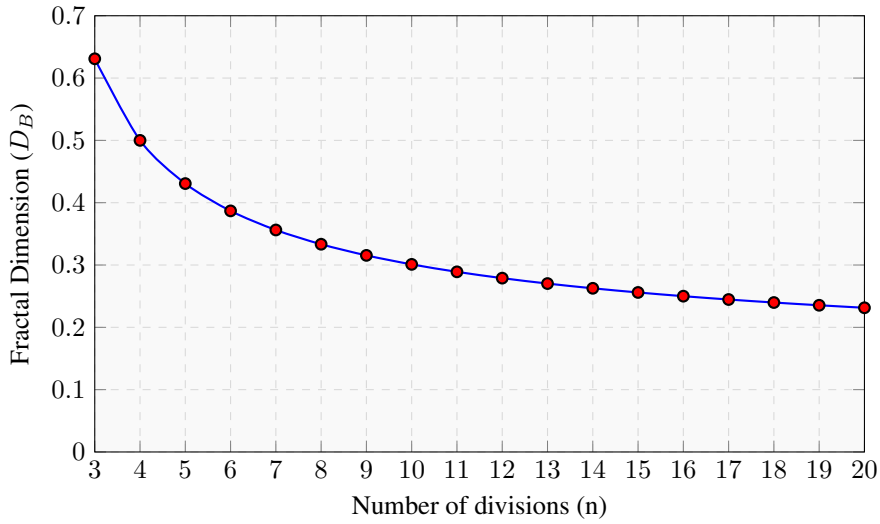
Since the derivative is negative, $D_B(C(\frac{n-2}{n}))$ is a decreasing function of n.

Furthermore, taking the limit as $n \to \infty$, we get

$$\lim_{n\to\infty} D_B(C(\frac{n-2}{n})) = \lim_{n\to\infty} \frac{\log 2}{\log n} = 0.$$

Thus, the dimension of $C(\frac{n-2}{n})$ decreases asymptotically to zero.

Remark 5.2: Theorem 5.1 shows that as n increases, the dimension of $C(\frac{n-2}{n})$ decreases asymptotically to 0. This indicates that the dimension is inversely proportional to the number of divisions n. The reason for this behavior is the increase in the length of the removed middle open interval. As n increases, the removed portion grows, and the dimension approaches zero, which corresponds to the topological dimension of a point.



Fractal dimension of $C\left(\frac{n-2}{n}\right)$ as a function of n

Theorem 5.2:

The sequence $\left\langle D_B\left(C\left(\frac{1}{n}\right)\right)\right\rangle_{n=3}^{\infty}$ of dimensions of the family of the Cantor set $\left\{C\left(\frac{1}{n}\right)\right\}$ is an increasing function of n, and it asymptotically approaches to unity.

For the set $C\left(\frac{1}{n}\right)$, we have:

$$D_B\left(C\left(\frac{1}{n}\right)\right) = \frac{\log 2}{\log\left(\frac{2n}{n-1}\right)} \quad \text{for} \quad n \geq 3$$

Differentiating D_B with respect to n:

$$\frac{d}{dn} D_B\left(C\left(\frac{1}{n}\right)\right) = \log 2 \cdot \frac{\left(-\frac{n-1}{2n}\right)\left(\frac{-2}{(n-1)^2}\right)}{\left[\log\left(\frac{2n}{n-1}\right)\right]^2} = \frac{\log 2}{n(n-1)\left[\log\left(\frac{2n}{n-1}\right)\right]^2}$$

Since $n \geq 3$, we have $n(n-1) > 0$ and $\log\left(\frac{2n}{n-1}\right) > 0$, hence:

$$\frac{d}{dn} D_B > 0$$

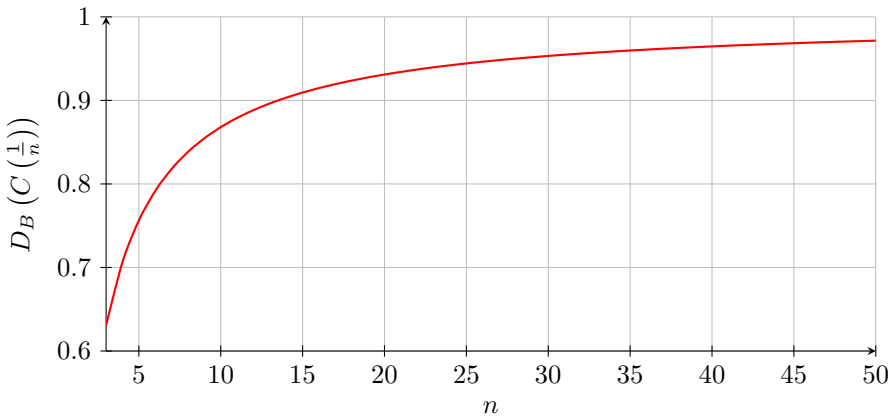
which shows D_B is increasing.

For the limit:

$$\lim_{n\to\infty} D_B\left(C\left(\frac{1}{n}\right)\right) = \lim_{n\to\infty} \frac{\log 2}{\log\left(\frac{2n}{n-1}\right)} = \frac{\log 2}{\log 2} = 1$$

Remark 5.3: Theorem 5.2 shows that as n increases, the length $\frac{1}{n}$ of the removed open interval

decreases, and the dimension of $C_{1/n}$ increases. That is, the length of the removed interval is inversely proportional to the dimension of $C_{1/n}$, and it asymptotically approaches unity, which is the topological dimension of the closed interval $[0, 1]$.



Dimension function $D_B\left(C\left(\frac{1}{n}\right)\right) = \dfrac{\log 2}{\log\left(\dfrac{2n}{n-1}\right)}$
showing the increasing behavior and asymptotic approach to 1.

Theorem 5.3: The sequence $\left\langle D_B\left(C\left(\frac{m}{n}\right)\right)\right\rangle_{m=1}^{\infty}$ of dimensions of the family of the Cantor set $\left\{C\left(\frac{m}{n}\right); m = 1, 2, 3, \ldots\right\}$ is an increasing function of m and asymptotically approaches unity.

Let us consider the function $f : [1, \infty) \to \mathbb{R}$ defined by

$$f(x) = \frac{\log(1+x)}{\log(1+2x)}.$$

Now, differentiating $f(x)$ with respect to x, we get

$$\frac{d}{dx}f(x) = \frac{(1+2x)\log(1+2x) - 2(1+x)\log(1+x)}{(1+x)(1+2x)[\log(1+2x)]^2} = \frac{h(x)}{g(x)}\text{(say)},$$

where $g(x) = (1+x)(1+2x)[\log(1+2x)]^2$ and $h(x) = (1+2x)\log(1+2x) - 2(1+x)\log(1+x)$.

Clearly, $g(x) > 0$ for all $x \in [1, \infty)$.

$$h(x) = \log(1{+}2x)^{1+2x} - \log(1{+}x)^{2(1+x)} = \log\left(\frac{1+2x}{1+x}\right)^{1+2x} - \log(1{+}x) = (1{+}2x)\log\left(1+\frac{x}{1+x}\right) - \log(1{+}x)$$

Now, $1 + x > \log(1 + x) \Rightarrow 1 + 2x > \log(1 + x)$ as $x \geq 1 \Rightarrow (1 + 2x)\log\left(1 + \frac{x}{1+x}\right) > \log(1 + x)$

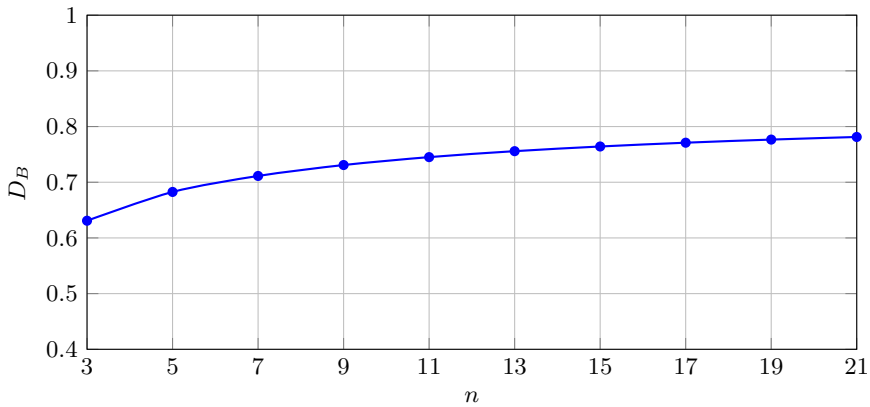
Thus, $h(x) > 0$, and hence, $\frac{d}{dx}f(x) > 0$ for all $x \in [1, \infty)$.

This shows that f is an increasing function of x, and hence, the dimension function $D_B\left(C\left(\frac{m}{n}\right)\right)$ is an increasing function of m.

Finally, let us consider the limiting case of $D_B\left(C\left(\frac{m}{n}\right)\right)$

$$\lim_{m\to\infty} D_B\left(C\left(\frac{m}{n}\right)\right) = \lim_{m\to\infty} \frac{\log(1+m)}{\log(1+2m)} = 1$$

Remark 5.4: Thus, if $C(\frac{m}{n})$ for $m = 1$ and $n = 2m+1$ is a sequence of Cantor sets, then the sequence of their dimensions $D_B(C(\frac{m}{n}))$, for $m = 1, 2, 3, \ldots$, is an increasing function of m. Moreover, this sequence of dimensions converges to 1, which is the topological dimension of the closed interval $[0, 1]$.



Dimension function $D_B\left(C\left(\frac{m}{n}\right)\right) = \dfrac{\log\left(1+\frac{n-1}{2}\right)}{\log n}$
where $n = 2m+1$ is odd, showing the increasing behavior and asymptotic approach to 1.

6 Application in Identifying Chaotic Nature in the Dynamics Produced by Piecewise Linear Maps

A map together with an initial seed value can produce a dynamics through an iteration scheme. For example, if we consider an iteration scheme like:

$$x_{n+1} = f(x_n) \tag{1}$$

where $f(x)$ is some well-defined map and we put an initial seed value of x (say x_0), the map can produce a dynamics where the trajectory will be formed by discrete points generated by repeated iteration of the map.

Though, in general, to have chaotic behavior in the dynamics produced this way, the map should be necessarily at least quadratic, it has been observed that piecewise linear maps like the tent map can produce qualitatively different kinds of behaviors and even can lead to chaotic or unpredictable situations in a bounded domain.

In studying this unpredictable behavior, the Cantor set plays a vital role. In many families of chaotic maps in one dimension, the ultimate attractor set is found to be a Cantor-like set with fractional dimension. dimension of the attractor set of the logistic map when the map shows chaotic behavior after attaining some critical value of the control parameter. The attractor in this case is a fractal and a Cantor-like set. In fact, fractals are the signature of chaos. As a consequence, it turns out that when we see self-similarity in the dynamics of a system of interest—that is, when the dynamics look the same at different scales of time and the attractor set has a fractional dimension, we can suspect deterministic chaos.

The following example justifies our claim:

6.1 Example 6.1

Consider the map $f : R \to R$ defined by:

$$f(x) = \begin{cases} 2kx, & x \leq \frac{1}{2}, \\ 2k(x-1)+1, & x > \frac{1}{2} \end{cases} \tag{2}$$

where $k > 0$ is a control parameter.

It is clear from the definition of f that $f^n(x) \to \infty$ as $n \to \infty$ for $x \in (-\infty, 0) \cup (1, \infty)$, where

$$f^n(x) = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}(x).$$

So, no interest is left out in this case as the trajectories formed by the dynamics of the map escape out to ∞.

But if we consider the dynamics produced by f for points in the interval $[0, 1]$, a completely different qualitative behavior emerges. The points of $[0, 1]$ which are mapped by f to either of the two intervals $(-\infty, 0)$ and $(1, \infty)$ will diverge to $-\infty$ or $+\infty$ under subsequent iteration of the map.

After the first iteration, the points in the set

$$F_1 = \left[0, \frac{1}{2k}\right] \cup \left[1 - \frac{1}{2k}, 1\right]$$

are mapped to $[0, 1]$, whereas the points in the middle open interval

$$D_1 = \left(\frac{1}{2k}, 1 - \frac{1}{2k}\right)$$

are mapped to $(-\infty, 0)$ or $(1, \infty)$, and so they diverge to ∞.

After the second iteration, the points in the set

$$D_2 = \left(\frac{1}{4k^2}, \frac{2k-1}{4k^2}\right) \cup \left(\frac{4k^2 - 2k + 1}{4k^2}, 1 - \frac{1}{4k^2}\right)$$

are mapped to D_1, and therefore, they will diverge to ∞.

The other points of F_1, i.e., the points in the set

$$F_2 = \left[0, \frac{1}{4k^2}\right] \cup \left[\frac{2k-1}{4k^2}, \frac{1}{2k}\right] \cup \left[1 - \frac{1}{2k}, \frac{4k^2 - 2k + 1}{4k^2}\right] \cup \left[1 - \frac{1}{4k^2}, 1\right]$$

are mapped to the set F_1.

This procedure gets repeated under further subsequent iterations. In view of this fact, it is easy to conclude that the attractor set A of f is nothing but the set $C(\frac{n-2}{n})$ where $n = 2k$. Moreover, we already showed that the set $C(\frac{n-2}{n})$ is a fractal, and hence, we can conclude that the map f exhibits chaotic behavior.

Below, we have shown the regular as well as chaotic behavior of the map with the variation in the parameter value with the help of the 'cobweb diagram' and 'bifurcation diagram,' which were generated

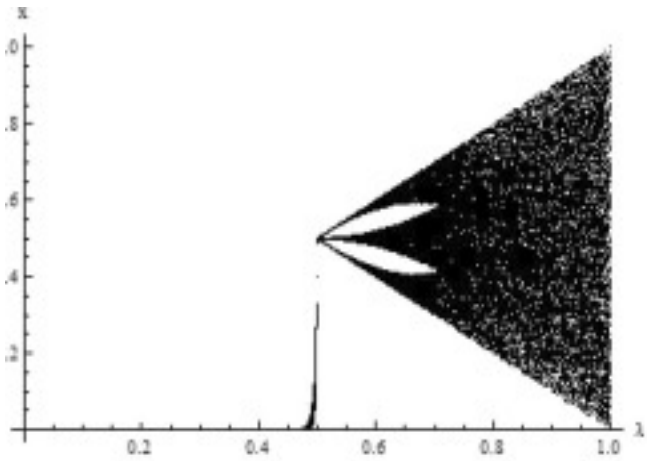


Figure 1:

by taking the data into account from repeated iteration of the map.

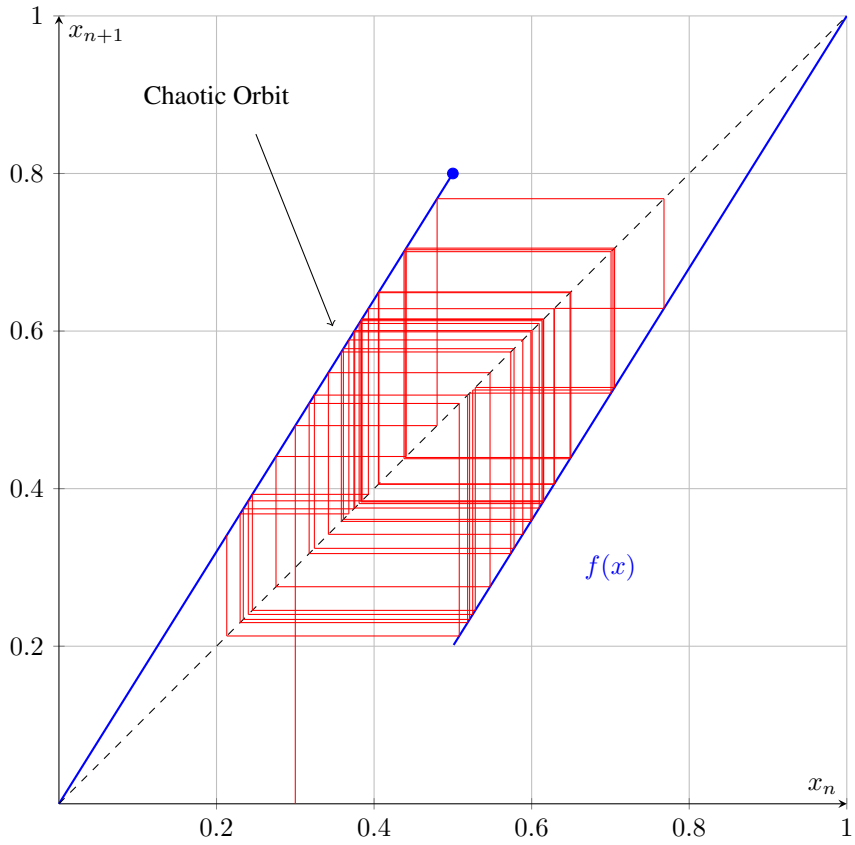
'Bifurcation' implies a qualitative change in the dynamics that takes place in a system with the variation in the parameter. A 'bifurcation diagram' is a diagram where the orbit points generated by the dynamics converge to, say, x, plotted against the parameter values. The bifurcation diagram helps us to visualize different types of dynamics that take place in a system with the variation in the parameter.

It is to be noted that the bifurcation diagram stops after $k = 1$, where a boundary crisis occurs, i.e., after $k = 1$, there exists an escape region in which if a trajectory lands, the next iteration rapidly takes the trajectory off toward $-\infty$.

The bifurcation diagram shows that after $k \approx 0.7$, chaos creeps into the dynamics produced by f, which is marked by areas having Cantor-like dust.

We have further verified the chaotic nature of the map with the help of a cobweb diagram as well. A 'cobweb diagram' is the geometrical representation of the iteration scheme by which the dynamics is produced. The following cobweb diagram was found for $k = 0.8$ by taking 300 iterations with the initial seed value $x_0 = 0.3$.

Cobweb Diagram ($k = 0.8$, $x_0 = 0.3$)



7 Conclusion

The primary aim of this dissertation was to extend the concept of the Cantor middle one-third set while preserving its embedding space, specifically the unit interval [0, 1]. We have identified three distinct methods for achieving this removal. Additionally, we have calculated the fractal dimensions of these generalized Cantor sets and examined their properties. From an application perspective, we have also explored the chaotic behavior of a piecewise linear map in relation to these generalized Cantor sets.

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